

Verification Plan Report

Project: AXI-Stream Master VIP Verification Plan

Version: 6.0

Verification Role: Master

1. UVM Environment Architecture

```
graph TD
    classDef test_layer fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,color:#7b1fa2;
    classDef test_obj fill:#f3e5f5,stroke:#7b1fa2,stroke-width:2px,stroke-dasharray: 5
5,color:#7b1fa2;
    classDef env_layer fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,color:#1565c0;
    classDef env_obj fill:#e3f2fd,stroke:#1565c0,stroke-width:2px,stroke-dasharray: 5
5,color:#1565c0;
    classDef agent_layer fill:#e0f2f1,stroke:#00695c,stroke-width:2px,color:#00695c;
    classDef agent_obj fill:#e0f2f1,stroke:#00695c,stroke-width:2px,stroke-dasharray: 5
5,color:#00695c;
    classDef dut_layer fill:#fff8e1,stroke:#f57f17,stroke-width:2px,color:#f57f17;

    subgraph UVM_Test_Layer [uvm_test]
        direction TB
        Test_Cfg[axi_test_config]:::test_obj
        Seq[axi_master_sequence]:::test_obj
        subgraph Env_Layer [uvm_env]
            direction TB
            Env_Cfg[axi_env_config]:::env_obj
            Scoreboard[axi_scoreboard]:::env_layer
            subgraph Agent_Layer [Master Agent - VIP Transmitter]
                direction TB
                Agent_Cfg[axi_agent_config]:::agent_obj
                Sequencer[axi_sequencer]:::agent_layer
                Driver["axi_master_driver (drives TVALID, TDATA, TLAST)"]:::agent_layer
                Monitor[axi_monitor]:::agent_layer
                Sequencer --> Driver
            end
        end
    end

    Seq -->|starts on| Sequencer
    subgraph Top_Level [Hardware Top]
        VIF[Virtual Interface]:::interface
        DUT["DUT (Slave - AXI Receiver)"]:::interface
        VIF <--> DUT
    end

    Driver --- VIF
    Monitor --- VIF
    Monitor -- "Analysis Port" --> Scoreboard
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class UVM_Test_Layer test_layer;  
class Env_Layer env_layer;  
class Agent_Layer agent_layer;  
class Top_Level dut_layer;  
class VIF,DUT dut_layer;
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2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake Protocol	TVALID-Driven / TREADY Back-Pressure Flow Control	The foundational conformance boundary of AXI-Stream. The VIP Master asserts TVALID and holds all payload signals stable until the DUT Slave asserts TREADY. The protocol mandates that a Transmitter (VIP) must NOT wait for TREADY before asserting TVALID, and once TVALID is HIGH it must not be deasserted until the handshake occurs. The DUT Slave is permitted to assert TREADY before or after TVALID, enabling all three canonical handshake orderings. (Section 2.2, p.18)
Packet Framing	TLAST Packet-Boundary Event Generation	TLAST is the sole packet-boundary indicator sourced from the VIP Master (Transmitter). The assertion count must be preserved between Transmitter and Receiver without exception. The VIP must correctly assert TLAST on the terminal beat of every packet and must not re-assert TLAST within a packet. Incorrect TLAST generation may corrupt DUT Slave state machines that use TLAST for buffer-drain triggering or arbitration change-points. (Section 2.6, p.26)

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Byte-Level Stream Semantics	TKEEP / TSTRB Qualifier-Driven Stream Typing	TKEEP qualifies bytes for network transport, enabling null byte insertion and removal. TSTRB distinguishes data bytes from position bytes within a sparse stream. The VIP must never drive the architecturally reserved encoding TKEEP=0/TSTRB=1. Correct qualifier encoding ensures the DUT Slave's data-reconstruction path receives an accurately typed byte stream. The scoreboard must independently reconstruct the expected stream and verify the DUT's output. (Section 2.5, Table 2-3, p.24-25)
Reset Sequencing	ARESETn Synchronous Deassertion with Transmitter TVALID Compliance	ARESETn may be asserted asynchronously at any time but must be deasserted synchronous to ACLK. During reset assertion, the VIP Master must unconditionally drive TVALID LOW. Furthermore, the first TVALID assertion post-reset is only architecturally valid at an ACLK rising edge that follows the rising edge of ARESETn. Mid-transfer reset injection by the test bench exercises the DUT Slave's ability to suppress in-flight data and return to IDLE without coherency corruption. (Section 2.8.2, p.28)
Wakeup Signaling (AXI5)	TWAKEUP Pre-Transfer Activity Indicator	The VIP Master (Transmitter) must assert TWAKEUP at least one ACLK cycle before TVALID. If TWAKEUP and TVALID are simultaneously HIGH, TWAKEUP must remain asserted until TREADY is seen. The DUT Slave is architecturally permitted to defer TREADY until TWAKEUP is observed, creating a potential deadlock condition if TWAKEUP is omitted. TWAKEUP must be glitch-free, generated from a register or glitch-free OR tree. (Section 2.3, p.20)

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Stream Identification	TID / TDEST Indexed Stream Multiplexing	TID provides per-beat stream identification enabling transfer-level interleaving across coexistent streams. TDEST provides coarse routing metadata for fabric-based stream switching. Transfers sharing TID and TDEST belong to the same stream; merging of transfers from different streams is prohibited. The VIP must maintain deterministic ordering within each TID/TDEST pair and never merge beats from different stream contexts. (Section 2.7, p.27; Section 4.2, p.41)
Parity Protection (AXI5)	Odd Parity Generation - Check_Type: Odd_Parity_Byte_All	When Check_Type=Odd_Parity_Byte_All, the VIP Master (Transmitter) must generate odd parity on all applicable check signals. Parity is driven on every cycle where the associated Check Enable condition is met. Critically, TDATACHK must be driven for ALL byte lanes including lanes where TKEEP=0 (null bytes). Single-bit error injection scenarios verify the DUT Slave's fault-detection mechanism. (Chapter 5, p.44-48)
Continuous Packets Mode	Continuous_Packets Protocol Constraint Enforcement	When Continuous_Packets=True, the VIP must enforce: absence of TSTRB, all TKEEP bits HIGH while TLAST=0, no TID/TDEST mutation mid-packet, and no inter-packet idle gaps. The protocol monitor must assert on any mid-packet TID change or null byte in this mode. This constraint ensures compatibility with Slave-side interconnect that relies on packet continuity for arbitration efficiency. (Section 3.3, p.38)

3. Test Requirements

Index	Name	Description
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REQ_MST_01	TVALID Stability and Handshake Integrity	The AXI-Stream handshake is predicated on the invariant that once a Transmitter asserts TVALID, it must hold TVALID HIGH and maintain all associated payload signals (TDATA, TSTRB, TKEEP, TLAST, TID, TDEST, TUSER) in a stable state until the handshake completes via coincident TREADY assertion. Critically, the Transmitter (VIP Master) is prohibited from conditioning TVALID assertion on a prior TREADY observation ? doing so would introduce a combinatorial dependency that risks deadlock. The only legal TVALID withdrawal point is after a completed handshake event. Verification Intent: The VIP Master driver must assert TVALID and hold all sideband signals with strict stability until TREADY is observed. Randomized protocol assertions on the monitor must fire on any mid-handshake TVALID deassertion or payload mutation. Negative-stimulus scenarios must inject TVALID drops before TREADY to characterize the DUT Slave's error response. All three handshake orderings (TVALID-first, TREADY-first, simultaneous) must be driven and binned separately within functional coverage.
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REQ_MST_02	Packet Boundary Generation via TLAST Assertion	TLAST is the exclusive packet-boundary indicator generated by the Transmitter (VIP Master). The number of TLAST assertions must be preserved end-to-end between Transmitter and Receiver without additional assertions (false boundaries) or suppressions (missed boundaries). The start of a packet is implicitly defined as the first transfer of a new TID/TDEST pair after reset, or the first transfer following the preceding packet's TLAST assertion for the same TID/TDEST. Merging of transfers from different packets (same or different TID/TDEST) is architecturally prohibited once TLAST has been observed. Verification Intent: VIP sequences must systematically cover: single-beat packets (TLAST=1 on beat 0), variable-length multi-beat packets (2 to 256 beats), and continuous back-to-back packet streams with no idle gap between TLAST=1 and the next TVALID=1. The scoreboard must independently reconstruct packet boundaries from the observed TLAST signal and verify that the DUT Slave correctly partitions the incoming byte stream at the reconstructed boundaries.
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REQ_MST_03	Byte Qualifier Conformance - TKEEP and TSTRB Encoding	TKEEP governs byte transport eligibility: when HIGH, the associated TDATA byte must be delivered to the destination; when LOW, the byte is a null byte eligible for removal. TSTRB qualifies TKEEP-asserted bytes as either data bytes (TSTRB=1) or positional placeholders (TSTRB=0). The combination TKEEP=LOW/TSTRB=HIGH is architecturally reserved and constitutes a protocol violation. The relative position of data bytes and position bytes within the stream must not change between Transmitter and Receiver, enforcing stream positional invariance. Verification Intent: The VIP Master must drive all valid TKEEP/TSTRB byte combinations under constrained-random stimulus, including fully-packed streams (all TKEEP=1, TSTRB=1), sparse configurations with positional bytes, and null-byte-bearing transfers. Dedicated negative-stimulus scenarios must inject the reserved encoding (TKEEP=0/TSTRB=1) on individual byte lanes. A real-time SVA assertion on the monitor must immediately flag this condition. For sparse stream coverage, the scoreboard must perform byte-level reconstruction and verify data-byte ordering invariance.
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REQ_MST_04	Null Transfer - Zero Data Bytes with TLAST Assertion	The AXI-Stream protocol permits a transfer in which TLAST is asserted but all TKEEP bits are deasserted, resulting in a packet that contains no data or position bytes. Such null transfers serve specific protocol-level functions: flushing buffers held in intermediate pipeline stages, signaling end-of-packet in absence of residual data, or completing an end-point operation expecting a terminal TLAST. A null transfer may be merged with an earlier transfer if both share TID/TDEST values and the earlier transfer has TLAST deasserted. Verification Intent: VIP sequences must generate null transfers in three contexts: (a) as the sole transfer forming an empty packet, (b) as the terminal transfer of a non-empty multi-beat packet, and (c) in back-to-back streams between two non-empty packets. The scoreboard must verify that the DUT Slave advances its packet-boundary state machine on each null TLAST event without incrementing any data-byte consumption counter.
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REQ_MST_05	Clock and Reset Protocol - TVALID Deassertion During ARESETn	ARESETn is an active-LOW global reset signal. It may be asserted asynchronously at any point in the interface timeline but must be deasserted synchronous to the rising edge of ACLK. The specification mandates that during ARESETn assertion, TVALID must be driven LOW. The Transmitter (VIP Master) must not assert TVALID until the first ACLK rising edge that follows the deassertion edge of ARESETn. Premature TVALID assertion post-reset constitutes a protocol violation that may corrupt the DUT Slave's initialization state. Verification Intent: Reset stimulus must be injected across three interface states: (a) idle with TVALID already LOW, (b) mid-packet with TVALID=1 and TREADY=0 (back-pressure active), and (c) at the precise handshake cycle where TVALID=TREADY=1. The VIP driver must drive TVALID LOW within one cycle of ARESETn assertion. Post-reset, compliance assertions must verify that the first TVALID rise occurs only on a valid post-reset ACLK edge. The DUT Slave's recovery behavior must be captured and scoreboarded.
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REQ_MST_06	AXI5 Parity Generation Across All Check Signals	The AXI5-Stream parity protection scheme employs odd parity across all interface signals when Check_Type=Odd_Parity_Byte_All. The VIP Master (Transmitter) must generate TVALIDCHK, TDATACHK, TLASTCHK, TIDCHK, TDESTCHK, and TUSERCHK on every cycle where the respective Check Enable condition is active. TDATACHK must be driven for all byte lanes unconditionally ? including lanes where TKEEP=0 ? as null-byte lanes still participate in parity computation. A check bit for a 1-bit signal (e.g., TVALIDCHK, TLASTCHK) equals the logical inversion of that signal per the critical control signal rule. Verification Intent: The scoreboard must independently compute expected parity for all check signals on every qualifying observation cycle and assert bit-exact equivalence against the monitored VIP outputs. Fault injection scenarios must introduce deliberate single-bit inversions on specific TDATACHK bit positions to exercise the DUT Slave's parity error-detection path. Coverage must bin each distinct check signal independently and confirm all byte positions of TDATACHK (including null-byte lanes) are observed.
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REQ_MST_07	TWAKEUP Pre-Transfer Wakeup Signaling Protocol (AXI5)	TWAKEUP is sourced by the Transmitter (VIP Master) and must be glitch-free to enable reliable asynchronous capture in a different clock domain, mandating generation from a register or glitch-free OR tree. The recommended assertion timing is at least one ACLK cycle prior to TVALID. When TWAKEUP and TVALID are simultaneously HIGH, TWAKEUP must remain asserted without interruption until TREADY is observed. The DUT Slave is architecturally permitted to gate TREADY pending TWAKEUP, creating a canonical deadlock scenario if TWAKEUP is never asserted while TVALID is HIGH. Verification Intent: Timing distributions to exercise: (a) TWAKEUP preceding TVALID by 1, 2, and 4 cycles; (b) simultaneous TWAKEUP and TVALID assertion; (c) TWAKEUP assertion followed by deassertion without a subsequent transfer. A deadlock watchdog timer must monitor for indefinite TREADY deferral. Coverage must bin all TWAKEUP lead-time variations and the simultaneous case explicitly.
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REQ_MST_08	Transfer Ordering and Multi-Stream Interleaving Integrity	The AXI-Stream protocol mandates strict intra-stream ordering: reordering of transfers within any given TID/TDEST stream is prohibited. Inter-stream interleaving at per-transfer granularity is permitted: transfers from distinct TID/TDEST streams may be interleaved on a beat-by-beat basis without restriction to TLAST boundaries. The VIP Master must maintain independent stream contexts for each active TID value and must not reorder beats within a shared stream context. Systems must also ensure that the total interleaving depth does not exceed the Receiver's capability, though the protocol does not define a specific maximum. Verification Intent: Multi-stream sequences must drive at least two concurrent TID streams with transfer-level interleaving at varying depths. The scoreboard must demultiplex monitored beats by TID/TDEST and independently verify per-stream ordering invariance. Protocol assertions must detect and flag any observable reordering event within a stream. Merging correctness must be verified: merges must only occur within matching TID/TDEST pairs where the earlier transfer has TLAST deasserted.
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4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	TVALID-First Handshake: VIP Master asserts TVALID with stable TDATA, TKEEP=0xF, TLAST=0. DUT Slave holds TREADY LOW for 1 to 16 randomized stall cycles then asserts TREADY. Monitor verifies TVALID and all sideband signals are stable across the entire stall window without a single-bit mutation.	REQ_MST_01

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TC_MST_002	TREADY-First Handshake: DUT Slave pre-asserts TREADY. VIP Master then asserts TVALID. Transfer must complete in that same clock cycle. Scoreboard verifies no pre-TVALID data sampling event is recorded.	REQ_MST_01
TC_MST_003	Negative - TVALID Drop Under Back-Pressure: VIP injects protocol violation by deasserting TVALID before TREADY is observed. Monitor SVA fires an immediate protocol error. DUT Slave behavior is logged to characterize its error response without deadlock.	REQ_MST_01
TC_MST_004	Single-Beat Packet: VIP drives TVALID=1, TLAST=1, TKEEP=0xFF simultaneously on beat 0. DUT Slave asserts TREADY. Scoreboard verifies TLAST event count increments by one and DUT returns to inter-packet idle state within one cycle of handshake completion.	REQ_MST_02
TC_MST_005	Multi-Beat Packet (N=2 to 256 randomized): VIP drives TVALID=1 continuously across N beats. TLAST=0 for beats 0 to N-2 and TLAST=1 on beat N-1. Randomized TREADY stalls injected. Scoreboard reconstructs the full packet and verifies data integrity.	REQ_MST_01, REQ_MST_02
TC_MST_006	Back-to-Back Packets: TVALID remains HIGH across the TLAST boundary. The final beat of packet N (TLAST=1) is immediately followed by the first beat of packet N+1 (TLAST=0) with zero idle cycles. Verify the DUT Slave partitions the two packets correctly and does not merge cross-boundary beats.	REQ_MST_02
TC_MST_007	Sparse Byte-Qualifier Stream: VIP drives constrained-random TKEEP and TSTRB per beat including null bytes (TKEEP=0), position bytes (TKEEP=1, TSTRB=0), and data bytes on a 4-byte bus. Scoreboard verifies byte ordering invariance. Monitor asserts that TKEEP=0/TSTRB=1 never appears.	REQ_MST_03
TC_MST_008	Reserved Qualifier Injection (Negative Stimulus): VIP injects TKEEP=0, TSTRB=1 on byte lane 2 of a single beat. Monitor SVA fires a protocol error. DUT Slave response is observed and logged. Verify system remains functional after the violation event.	REQ_MST_03
TC_MST_009	Null Transfer at Packet End: VIP drives a 4-beat data packet followed immediately by a null beat (TVALID=1, TLAST=1, TKEEP=0x0). Scoreboard verifies DUT advances packet-boundary state without crediting data bytes to the null transfer event.	REQ_MST_04

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TC_MST_010	Standalone Null Packet: VIP drives TVALID=1, TLAST=1, TKEEP=0x0 as the first and only transfer of a packet. DUT must complete handshake. Scoreboard verifies zero data-byte consumption and correct transition to inter-packet idle.	REQ_MST_04
TC_MST_011	Mid-Transfer Reset: ARESETn driven LOW while VIP has TVALID=1 and DUT TREADY=0. VIP must deassert TVALID within one ACLK cycle. After ARESETn deassertion, verify TVALID is only reasserted at the first valid post-reset ACLK edge.	REQ_MST_05
TC_MST_012	Reset at Handshake Boundary: ARESETn asserted at the exact ACLK edge where TVALID=TREADY=1. Verify the transfer is neither partially sampled nor silently dropped. DUT Slave must return to IDLE and the post-reset first transfer must be correct.	REQ_MST_05
TC_MST_013	AXI5 Parity - Nominal Traffic: VIP drives correct odd parity on TVALIDCHK, TDATACHK, and TLASTCHK for all beats across a 2000-transfer regression run. Scoreboard performs bit-exact independent parity computation and asserts equivalence on every observed cycle.	REQ_MST_06
TC_MST_014	AXI5 Parity - Null Byte Lane Coverage: With TKEEP=0 on byte lanes 1 and 3 of a 4-byte bus, VIP must still generate correct TDATACHK for those inactive lanes. Scoreboard verifies parity is computed for all byte positions regardless of TKEEP state.	REQ_MST_06
TC_MST_015	AXI5 Parity Fault Injection: VIP deliberately inverts bit 1 of TDATACHK on transfer index 99. Verify DUT Slave activates its error-detection path (error flag, interrupt, or protocol assertion) within the cycle of the parity anomaly.	REQ_MST_06
TC_MST_016	TWAKEUP - Recommended Timing: VIP asserts TWAKEUP exactly 2 cycles before TVALID. Verify DUT Slave accepts TREADY assertion within the wakeup latency budget and no deadlock occurs within a 50-cycle watchdog window.	REQ_MST_07
TC_MST_017	Multi-Stream Interleaving: VIP drives two streams (TID=0, TID=1) interleaved at transfer granularity across 64 beats (alternating TID per beat). Scoreboard demultiplexes by TID and independently verifies per-stream ordering invariance and absence of cross-stream data contamination.	REQ_MST_08

5. Functional Coverage Groups

Group Name	Description
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cg_handshake_ordering	Three-bin enumerated coverage of handshake ordering: TVALID-first, TREADY-first, simultaneous. Crossed with TREADY stall duration bins: 0, 1-4, 5-16, 17-32 cycles. Must achieve closure across all 12 cross-product cells.
cg_packet_size_distribution	Packet-length bins: 1 beat, 2-4 beats, 5-32 beats, 33-256 beats. Separate bins for back-to-back and isolated packet categories. Cross with TREADY stall presence (0 stalls vs at least one stall).
cg_keep_tstrb_combinations	Per-byte lane encoding bins: data byte (TKEEP=1/TSTRB=1), position byte (TKEEP=1/TSTRB=0), null byte (TKEEP=0/TSTRB=0). Negative case (TKEEP=0/TSTRB=1) must appear in negative-stimulus run only (zero coverage in functional runs).
cg_null_transfer_context	Null transfer occurrence context bins: standalone null packet, terminal null after data packet, between two data packets. Cross with packet length of the preceding data packet (0, 1-4, 5+ beats).
cg_reset_injection_state	Reset injection interface-state bins: (a) idle TVALID=0, (b) mid-packet TVALID=1/TREADY=0, (c) handshake coincidence TVALID=TREADY=1. Cross with ARESETn assertion duration: 1, 2-4, 5+ cycles.
cg_parity_check_coverage	Per-byte-lane parity check coverage for TDATACHK on 4-byte bus. Active lane bins (TKEEP=1) and inactive lane bins (TKEEP=0) for bytes 0-3. Separate fault-injection bins: individual bit inversions on TDATACHK[3:0].
cg_twakeup_lead_time	TWAKEUP lead-time relative to TVALID rising edge. Bins: -1 (post-TVALID, violation scenario), 0 (simultaneous), 1 cycle, 2-4 cycles, 5+ cycles. Cross with TREADY response latency post-TWAKEUP.
cg_stream_interleaving	Active TID stream count bins: 1, 2, 4. Interleaving granularity (consecutive transfers per TID before stream switch): 1, 2-4, 5-8, 9+. Cross verifies per-stream ordering at maximum interleaving depth.

6. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input (to VIP Master and DUT Slave)	1	Global synchronization clock. All input signals are sampled on the rising edge of ACLK. All output signal changes must occur after the rising edge of ACLK. (Section 2.8.1, p.28)
ARESETn	Input (to VIP Master and DUT Slave)	1	Active-LOW global reset. Asynchronous assertion, synchronous deassertion on ACLK rising edge. Mandates TVALID=0 from the VIP Master (Transmitter) for its entire assertion duration. (Section 2.8.2, p.28)

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TVALID	Output (VIP Master / Transmitter)	1	Driven by the VIP Master. Indicates a valid transfer payload is present. Once asserted, must remain HIGH until TREADY is observed. The VIP must not condition TVALID on a prior TREADY observation. (Section 2.2, p.18)
TREADY	Input (from DUT Slave / Receiver)	1	Driven by the DUT Slave to signal readiness to accept a transfer. The Slave may assert TREADY before or after TVALID. Default value is HIGH if the signal is absent. (Section 3.1.1, p.34)
TDATA	Output (VIP Master / Transmitter)	TDATA_WIDTH (integer x8 bits, recommended 8-1024 bits)	Primary data payload. Low-order bytes represent earlier stream bytes. Must remain stable for the full TVALID assertion window prior to handshake completion. (Section 2.4.1, p.21)
TSTRB	Output (VIP Master / Transmitter)	TDATA_WIDTH /8	Per-byte qualifier. When HIGH (with TKEEP=1), identifies the associated byte as a data byte. When LOW (with TKEEP=1), identifies the byte as a positional placeholder. (Section 2.5.2, Table 2-3, p.24-25)
TKEEP	Output (VIP Master / Transmitter)	TDATA_WIDTH /8	Per-byte null-byte qualifier. HIGH indicates the byte must be transported to the destination. LOW indicates a null byte eligible for removal. Default: all-HIGH if absent from interface. (Section 2.5.1, p.24)
TLAST	Output (VIP Master / Transmitter)	1	Packet-boundary indicator generated by VIP Master. Asserted on the terminal beat of each packet. TLAST assertion count must be preserved without suppression or duplication between Transmitter and Receiver. (Section 2.6, p.26)
TID	Output (VIP Master / Transmitter)	TID_WIDTH (recommended <= 8 bits)	Stream identifier enabling per-beat multiplexing of independent data streams. All transfers sharing TID and TDEST values belong to the same stream and must be delivered in order. (Section 2.7, p.27)
TDEST	Output (VIP Master / Transmitter)	TDEST_WIDTH (recommended <= 8 bits)	Routing identifier providing coarse destination addressing for stream-level routing. Merging of transfers with differing TID or TDEST values is prohibited. (Section 2.7, p.27)
TUSER	Output (VIP Master / Transmitter)	TUSER_WIDTH (recommended: integer multiple of TDATA_WIDTH /8)	User-defined sideband signal, byte-aligned to TDATA. For byte x: TUSER[((x*m)+(m-1)):(x*m)]. Transfer for TUSER bits is not required or guaranteed for bytes where TKEEP=LOW. (Section 2.9, p.29)

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TWAKEUP	Output (VIP Master Transmitter, AXI5-Stream only)	1	Activity indicator for power-domain management. Must be glitch-free. Recommended assertion at least one ACLK cycle before TVALID. Must remain HIGH until TREADY when simultaneously asserted with TVALID. (Section 2.3, p.20)
TVALIDCHK	Output (VIP Master Transmitter, AXI5-Stream only)	1	Odd parity check bit for TVALID. Check Enable: ARESETn=1. Equal to the logical inversion of TVALID under the critical control signal odd-parity rule. (Table 5-1, p.48)
TDATACHK	Output (VIP Master Transmitter, AXI5-Stream only)	TDATA_WIDTH / 8	Odd parity check per 8-bit byte lane of TDATA. Check Enable: TVALID=1. Must be driven for ALL byte lanes including TKEEP=0 lanes. Bit n covers TDATA[(8n+7):8n]. (Table 5-1, p.48; Section 5.3, p.46)
TLASTCHK	Output (VIP Master Transmitter, AXI5-Stream only)	1	Odd parity check bit for TLAST. Check Enable: TVALID=1. Equals the logical inversion of TLAST per the critical control signal odd-parity rule. (Table 5-1, p.48)
TREADYCHK	Input (from DUT Slave Receiver, AXI5-Stream only)	1	Odd parity check bit for TREADY, generated by the DUT Slave. Check Enable: ARESETn=1. Scoreboard independently verifies this DUT output against the expected odd parity of TREADY. (Table 5-1, p.48)

7. Verification Environment Structure

Root: axi_stream_master_vip_tb/

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|-- master_agent : VIP Master Agent: Sequencer, Driver (drives TVALID/TDATA/TLAST/parity), Monitor
|-- env : UVM Environment: Scoreboard, Protocol Checker, Coverage Collector, EnvConfig
|-- sequences : Stimulus library: handshake variants, sparse-stream, back-to-back, null-packet, parity-gen, parity-inject, reset, i
|-- tests : UVM test classes mapping each scenario to sequence combinations and config overrides
|-- assertions : SVA property file for real-time protocol conformance (stability, reserved-encoding, parity)
|-- top : Simulation top: DUT (Slave) instantiation, virtual interface binding, clock and reset generation
|-- verif_plan : Artifacts: YAML traceability matrix, PDF report, sub-prompt reference files

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